

ROY GONZALEZ

Principal Architect · Compilers, DSLs & Neuro-Symbolic AI · Distributed Systems

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PROFESSIONAL SUMMARY

Software architect and engineer with 20 years designing and building enterprise platforms — mission-critical trading systems, logistics integration and cloud-hosted enterprise products — and founder of **MLambda**, a compiler, DSL and artificial-intelligence company that delivers **neuro-symbolic solutions**. MLambda designs and builds programming languages, compilers and runtimes: native code through **LLVM**, or DSLs embedded in the **C#/.NET, Java, Go and Rust** ecosystems a client already runs. Deep expertise in distributed systems, the actor model, DDD and formal verification, with a record of setting architectural standards, reducing technical debt, leading CI/CD adoption and mentoring engineers across US and Latin American teams.

CORE COMPETENCIES

Compilers & DSLs	Language & compiler design · LLVM IR & native code generation · DSL design & tooling · Parser combinators, EBNF & typed ASTs · Type systems (Hindley–Milner) · Language runtimes · FFI · WebAssembly
Neuro-Symbolic AI	Symbolic reasoning over LLMs · LLM agents & RAG · Transformers (PyTorch) · Modal & temporal logic · Formal specification & model checking (TLA+, Z notation)
Architecture	Enterprise Architecture (TOGAF) · Distributed Systems · Microservices · Domain-Driven Design · Event-Driven & Actor-Model Architectures · Messaging · API Integration
Platform & Cloud	Kubernetes · Helm · Docker · Terraform · AWS · CI/CD · High Availability & Fault Tolerance · Observability
Languages & Data	C# / .NET · Go · Rust · C · Java · Python · JavaScript / TypeScript / React · SQL Server · PostgreSQL · MongoDB · Oracle · Apache Spark
Leadership	Technical mentoring · Architecture governance · Tech-debt reduction · Scrum / Agile · TDD / BDD coaching

PROFESSIONAL EXPERIENCE

Founder & Principal Architect | MLambda

2019 – Present

Heredia, Costa Rica — Compilers, domain-specific languages and neuro-symbolic artificial intelligence

- Founded and lead a compiler, DSL and AI company that builds **neuro-symbolic solutions**: systems where neural networks acquire knowledge and a symbolic reasoner guarantees it, so results are verified rather than guessed.
- Design and build **custom programming languages, compilers and runtimes** for clients — lowering to native code through **LLVM**, or delivering DSLs, interpreters and code generators hosted in **C#/.NET, Java, Go or Rust**.
- Aleph** — designed and implemented a functional programming language end to end: grammar, type checker, LLVM-based native code generation, standard library and a native FFI to C, assembly, LLVM IR and WebAssembly.
- Genesis** — a compiler workbench for building DSLs (published on NuGet): grammars become executable parsers, schemas become typed syntax trees, and specifications become TLA+ model-checked properties.
- Turing** — an architecture platform that turns a plain-language description into a formal domain model, a model-checked specification and a generated, tested, containerised system, using LLM agents under symbolic rules.
- Hilbert** — research into a neuro-symbolic transformer that reasons across first-order, relational, modal, temporal and sortal logic, with fast weights that learn from each input without retraining.
- Actors & Data** — author of MLambda.Actors, an MIT-licensed actor runtime for .NET with supervision, clustering, gossip membership and mTLS (NuGet); and of an actor-native distributed database whose consistency model is a checked TLA+ specification.

Stack: C#/.NET 10 · LLVM · Rust · C · Python · PyTorch · TLA+ · Z notation · Kubernetes · Helm

Software Developer | QAT Global

Jul 2025 – Present

Costa Rica · Remote — Nearshore custom software development for US enterprise clients

- Deliver custom enterprise software in C# and .NET for US clients, across design and implementation.

Software Developer | EagleView

Aug 2023 – Jun 2025

United States · Remote — 3D modeling, CAD and GIS platform

- Built CAD tooling for 3D architectural modeling — extrusion, scaling, structure creation — with JavaScript, Python and Three.js, integrating models into GIS environments via MapView.
- Optimised geometry-processing workflows and the interaction design of complex interactive 3D tools.

Principal Software Engineer | TradeStation

Jan 2020 – Jun 2023

Costa Rica · Remote — Mission-critical online trading platform, US fintech

- Designed and evolved the microservices behind a high-availability trading platform in .NET, Go and Kubernetes, where reliability, low latency and fault tolerance were the governing architectural constraints.
- Built broker and cryptocurrency APIs (bars, positions, order placement) and led feature architecture at production scale, with Helm-based deployment and performance tooling that improved service response times.

Stack: C# · Go · Kubernetes · Helm · Terraform · AWS · GraphQL · PostgreSQL · MongoDB

Software Developer | Mobilize.Net

Nov 2016 – Jul 2019

Costa Rica — Automated code-migration products (source-to-source translation)

- Redesigned the front and back end of the WebMAP migration product, simplifying it and improving performance; helped teams adopt TDD, simpler CI/CD and profiling discipline.

Software Architect | Aurea Software

Jun 2015 – Nov 2016

United States · Remote — Enterprise software at cloud-hosted scale

- Architected the migration from SQL Server Integration and Analysis Services to Mondrian and Apache Spark, modernising the analytics platform.
- Remediated technical debt across enterprise products, led the performance programme, and added a CI job that profiled every build for bottlenecks and memory leaks; mentored teams on testing and code standards.

Software Architect | Accenture

Sep 2013 – Jun 2015

Heredia, Costa Rica — Enterprise consulting for health-sector clients

- Designed a modular EPiServer architecture with a reusable component library shared across multiple sites.
- Applied TOGAF to unify business, software and infrastructure views, and introduced Scrum, SOLID, TDD, continuous integration and Octopus deployment; contributed to the engagement that won the client a major multi-year contract.

Software Engineer | Isthmus (DHL account)

Jan 2010 – Sep 2013

Costa Rica — Logistics integration systems

- Built DHL integration systems on publish-subscribe patterns and WCF, progressing from team lead to architect.
- Performed business and data analysis (Analysis Services) on DHL's logistics core for reliability-critical solutions.

Software Engineer / Architect | GPI Consultores · GPS Satellite

Mar 2006 – Jan 2010

Costa Rica — Logistics, banking and GPS telematics

- Built distributed systems integrating GPS devices over TCP/IP and UDP with real-time data processing.
- Developed single sign-on middleware, DHL trade and tracking applications and e-commerce systems; led the maintenance team for a Citibank credit-card platform and trained teams on BizTalk Server.

EDUCATION

BS, Computer Software Engineering · Universidad Nacional, Costa Rica

2002 – 2006

Bachelor's, Philosophy of Formal Science · Universidad Nacional, Costa Rica

2002 – 2006

LANGUAGES

Spanish (native) · English (advanced — 10+ years working remotely with US teams)

LINKS

MLambda	www.mlambda.net	MLambda.AI	www.mlambda.net/MLambda.AI — learn AI with Hilbert
Genesis	genesis.mlambda.net — DSL workbench	Actors	actors.mlambda.net — .NET actor runtime
GitHub	github.com/mlambda-net	Personal site	yordivad.github.io